Lab Eleven – Point Estimation and Resampling

Tips

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.
- Make sure to use "enter" to make your code readable, so it doesn't run off the page. I switched the Quarto document to automatically render to .pdf, so you can see what you're submitting by default. You can switch back for the highlighting by moving the html block above the pdf block in the YAML header.
- Don't forget that you can copy-and-paste code when you're doing something similar as we did in class!
- Make sure to plot all computed probabilities. This is good `ggplot` practice AND will help make sure you don't make a mistake when computing the probability.

1 Introduction

When performing point estimation, there isn't just one *right* answer. For example, the Method of Moments (MOM) and Maximum Likelihood Estimates (MLEs) aren't always the same.

One question that comes up, is why we calculate the sample variance as

$$S_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

when it is supposed to be the "average squared distance from the mean, i.e.,

$$S_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

2 Comparing Estimators

2.1 Initial Simulation

In this lab, we will explore why we use S_{n-1}^2 and not S_n^2. To do this, we will generate data from four known distributions:

- Poisson($\lambda = 2$)
- Binomial$\left(n = 50, p = \frac{5-\sqrt{21}}{10}\right)$
- Normal($\mu = 0, \sigma = \sqrt{2}$)
- Exponential$\left(\lambda = \frac{1}{\sqrt{2}}\right)$.

Remark: Note that I have selected the parameters so that the true variance for all four models is 2.

For each distribution, write a loop that completes the following 1000 times. Put `set.seed(7272)` at the top of your code chunk.

- Generate a sample of size n from the distribution.
- Calculate and save S_{n-1}^2 and S_n^2.

Start with $n = 20$. Is there evidence that S_{n-1}^2 is a better estimator than S_n^2?

Create a plot or combination of plots (using patchwork) to show the results across all four distributions. Make sure to create a corresponding table that numerically summarizes the results of the simulation. You should include

$$\text{Bias} = \text{Average of Estimates} - \text{True Value}$$
$$\text{Precision} = \frac{1}{\text{Variance of Estimates}}$$
$$\text{Mean Square Error} = \text{Bias}^2 + \text{Variance of Estimates.}$$

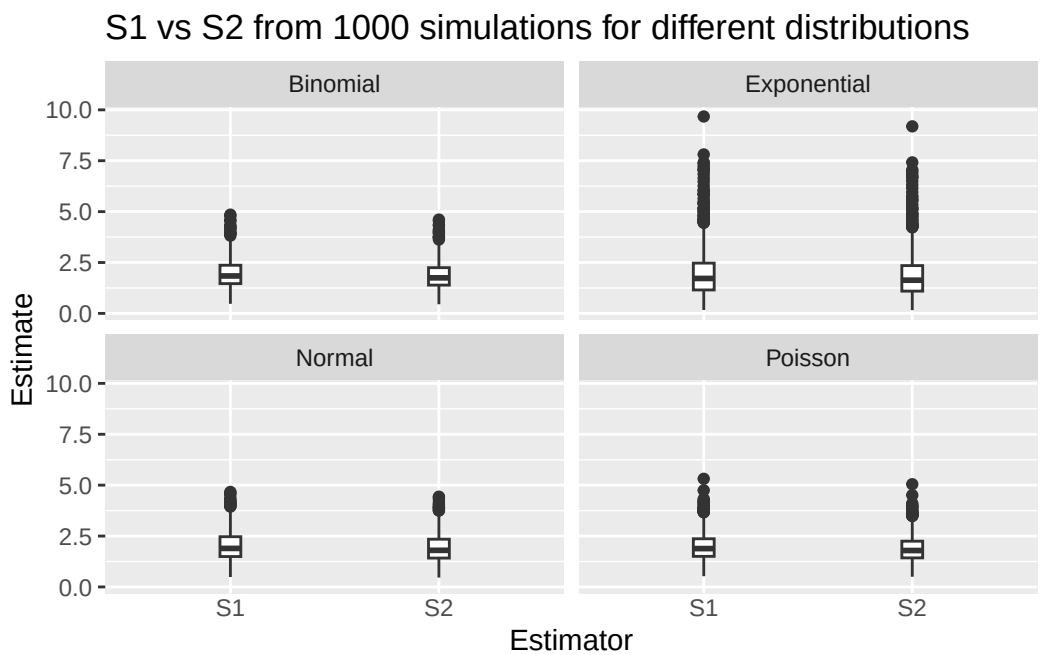
Solution

There is not enough evidence that S_{n-1}^2 is a better estimator than S_n^2 because across all distributions, on average (across 1000 simulations), S_{n-1}^2 has lower bias compared to S_n^2, but also lower precision.

```r
set.seed(7272)

n.sims <- 1000
# sample size is 20
n <- 20

# Poisson
S1_pois <- rep(NA, n.sims)
S2_pois <- rep(NA, n.sims)

for(i in 1:n.sims){
  curr.sample <- rpois(n=n, lambda = 2)
  x_bar = mean(curr.sample)
  # don't need for (i in 1:n){Xi <- curr.sample[i] S1[i] <- (1/(n - 1)) * sum((Xi - x_bar)^2) S2[i] <- 1/n
  S1_pois[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
  S2_pois[i] <- sum((curr.sample - x_bar)^2) / n
  }

# Binomial
p <- (5-sqrt(21))/10

S1_binom <- rep(NA, n.sims)
S2_binom <- rep(NA, n.sims)

for(i in 1:n.sims){
  curr.sample <- rbinom(n = n, size = 50, prob = p)
  x_bar <- mean(curr.sample)
  S1_binom[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
  S2_binom[i] <- sum((curr.sample - x_bar)^2) / n
}

# Normal
S1_norm <- rep(NA, n.sims)
S2_norm <- rep(NA, n.sims)

for(i in 1:n.sims){
  curr.sample <- rnorm(n = n, mean = 0, sd = sqrt(2))
  x_bar <- mean(curr.sample)
  S1_norm[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
  S2_norm[i] <- sum((curr.sample - x_bar)^2) / n
}

# Exponential
S1_exp <- rep(NA, n.sims)
S2_exp <- rep(NA, n.sims)
```

```r

for(i in 1:n.sims){
  curr.sample <- rexp(n = n, rate = 1/sqrt(2))
  x_bar <- mean(curr.sample)
  S1_exp[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
  S2_exp[i] <- sum((curr.sample - x_bar)^2) / n
}
```

```r
raw <- tibble(
  Estimator= rep(c("S1", "S2"), each = 1000, times = 4),
  Distribution= rep(c("Poisson", "Binomial", "Normal", "Exponential"),
                    each = 2000),
  Estimate=c(S1_pois, S2_pois, S1_binom, S2_binom,
             S1_norm, S2_norm, S1_exp, S2_exp)
)

ggplot(raw) +
  geom_boxplot(aes(x = Estimator,
                   y= Estimate),
               width = 0.1) +
  ggtitle("S1 vs S2 from 1000 simulations for different distributions") +
  theme(axis.ticks.x = element_blank()) +
  facet_wrap(~ Distribution)
```




```r
# results.S1 <- tibble(
#   Distribution = c("Poisson", "Binomial", "Normal", "Exponential"),
#   Bias = c(
#     mean(S1_pois) - 2,
#     mean(S1_binom) - 2,
#     mean(S1_norm) - 2,
#     mean(S1_exp) - 2
#   ),
#   Variance = c(
#     var(S1_pois),
#     var(S1_binom),
#     var(S1_norm),
```

```r
#     var(S1_exp)
#   ),
#   Precision = c(
#     1/var(S1_pois),
#     1/var(S1_bin),
#     1/var(S1_norm),
#     1/var(S1_exp)
#   ),
#   MSE = c(
#     (mean(S1_pois) - 2)^2 + var(S1_pois),
#     (mean(S1_bin) - 2)^2 + var(S1_bin),
#     (mean(S1_norm) - 2)^2 + var(S1_norm),
#     (mean(S1_exp) - 2)^2 + var(S1_exp)
#   )
# )

results <- tibble(
  Distribution = rep(c("Poisson", "Binomial", "Normal", "Exponential"), each = 2),
  Estimator = rep(c("S1", "S2"), times = 4),
  Bias = c(
    mean(S1_pois) - 2, mean(S2_pois) - 2,
    mean(S1_binom)  - 2, mean(S2_binom)  - 2,
    mean(S1_norm) - 2, mean(S2_norm) - 2,
    mean(S1_exp)  - 2, mean(S2_exp)  - 2
  ),
  Variance = c(
    var(S1_pois), var(S2_pois),
    var(S1_binom),  var(S2_binom),
    var(S1_norm), var(S2_norm),
    var(S1_exp),  var(S2_exp)
  ),
  Precision = 1 / Variance,
  MSE = Bias^2 + Variance
)
```

```r
# ggplot(results, aes(x = Estimator, y = Bias, fill = Estimator)) +
#   geom_col() +
#   facet_wrap(~ Distribution) +
#   labs(title = "Bias")
```

Distribution	Estimator	Bias	Variance	Precision	MSE
Poisson	S1	-0.02	0.48	2.10	0.48
Poisson	S2	-0.12	0.43	2.33	0.44
Binomial	S1	-0.04	0.47	2.15	0.47
Binomial	S2	-0.13	0.42	2.38	0.44
Normal	S1	0.00	0.49	2.03	0.49
Normal	S2	-0.10	0.44	2.25	0.45
Exponential	S1	-0.02	1.43	0.70	1.43
Exponential	S2	-0.12	1.29	0.77	1.30

2.2 Simulation over various sample sizes

Consider the effect of the sample size have on these results. Provide a graphical and numerical summary of what happens as n increases – say $n = 20$, $n = 50$, $n = 100$, and $n = 500$. Does the result in your initial simulation hold across samples? Or, do things change as the sample size changes? Ensure to set your randomization seed.

Solution

As sample size increases, the estimates of variance become less variable, and the average values of S1 and S2 seem to converge. In the original simulation, the samples of S1 and S2 for Exponential and Poisson contain a lot of outliers and their distributions seem right-skewed, but they become approximately Gaussian at n=100 and n=500.

- Poisson

```r
set.seed(7272)

n.sims <- 1000
ss <- c(20, 50, 100, 500)

S1_pois <- list()
S2_pois <- list()

for(n in ss){
  S1 <- rep(NA, n.sims)
  S2 <- rep(NA, n.sims)
  for(i in 1:n.sims){
    curr.sample <- rpois(n = n, lambda = 2)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  # convert number to string to use as a name
  S1_pois[[as.character(n)]] <- S1
  S2_pois[[as.character(n)]] <- S2
}

# raw data
raw_pois <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  # so that they're in order
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_pois[["20"]], S2_pois[["20"]],
               S1_pois[["50"]], S2_pois[["50"]],
               S1_pois[["100"]], S2_pois[["100"]],
               S1_pois[["500"]], S2_pois[["500"]])
)
# summary
results_pois <- tibble(
  n = rep(c(20, 50, 100, 500), each=2),
  Estimator = rep(c("S1", "S2"), times = 4),
  Bias = c(
    mean(S1_pois[["20"]]) - 2, mean(S2_pois[["20"]]) - 2,
    mean(S1_pois[["50"]]) - 2, mean(S2_pois[["50"]]) - 2,
    mean(S1_pois[["100"]]) - 2, mean(S2_pois[["100"]]) - 2,
    mean(S1_pois[["500"]]) - 2, mean(S2_pois[["500"]]) - 2
  ),
  Variance = c(
    var(S1_pois[["20"]]),  var(S2_pois[["20"]]),
    var(S1_pois[["50"]]),  var(S2_pois[["50"]]),
    var(S1_pois[["100"]]), var(S2_pois[["100"]]),
    var(S1_pois[["500"]]), var(S2_pois[["500"]])
  ),
  Precision = 1 / Variance,
  MSE = Bias^2 + Variance)

```

```r
# table
results_pois |> knitr::kable(digits = 2)
```

n	Estimator	Bias	Variance	Precision	MSE
20	S1	-0.02	0.48	2.10	0.48
20	S2	-0.12	0.43	2.33	0.44
50	S1	-0.03	0.20	5.03	0.20
50	S2	-0.07	0.19	5.24	0.20
100	S1	0.01	0.11	9.34	0.11
100	S2	-0.01	0.10	9.53	0.11
500	S1	0.00	0.02	52.26	0.02
500	S2	-0.01	0.02	52.47	0.02

```r
# plot
p_pois2 = ggplot(raw_pois) +
  geom_boxplot(aes(x = Estimator,
                   y= Estimate),
               width = 0.1) +
  ggtitle("Poisson") +
  theme(axis.ticks.x = element_blank()) +
  facet_wrap(~ n)
```

- Binomial

```r
set.seed(7272)

S1_binom <- list()
S2_binom <- list()

for(n in ss){
  S1 <- rep(NA, n.sims)
  S2 <- rep(NA, n.sims)
  for(i in 1:n.sims){
    curr.sample <- curr.sample <- rbinom(n = n, size = 50, prob = p)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_binom[[as.character(n)]] <- S1
  S2_binom[[as.character(n)]] <- S2
}

# raw data
raw_binom <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_binom[["20"]], S2_binom[["20"]],
               S1_binom[["50"]], S2_binom[["50"]],
               S1_binom[["100"]], S2_binom[["100"]],
               S1_binom[["500"]], S2_binom[["500"]])
)
# summary
results_binom <- tibble(
  n = rep(c(20, 50, 100, 500), each=2),
  Estimator = rep(c("S1", "S2"), times = 4),
```

```r
  Bias = c(
    mean(S1_binom[["20"]]) - 2, mean(S2_binom[["20"]]) - 2,
    mean(S1_binom[["50"]]) - 2, mean(S2_binom[["50"]]) - 2,
    mean(S1_binom[["100"]]) - 2, mean(S2_binom[["100"]]) - 2,
    mean(S1_binom[["500"]]) - 2, mean(S2_binom[["500"]]) - 2
  ),
  Variance = c(
    var(S1_binom[["20"]]),  var(S2_binom[["20"]]),
    var(S1_binom[["50"]]),  var(S2_binom[["50"]]),
    var(S1_binom[["100"]]), var(S2_binom[["100"]]),
    var(S1_binom[["500"]]), var(S2_binom[["500"]])
  ),
  Precision = 1 / Variance,
  MSE = Bias^2 + Variance)

# table
results_binom |> knitr::kable(digits = 2)
```

n	Estimator	Bias	Variance	Precision	MSE
20	S1	-0.01	0.47	2.13	0.47
20	S2	-0.11	0.42	2.36	0.44
50	S1	-0.03	0.19	5.26	0.19
50	S2	-0.07	0.18	5.48	0.19
100	S1	0.00	0.10	9.67	0.10
100	S2	-0.02	0.10	9.86	0.10
500	S1	0.00	0.02	53.65	0.02
500	S2	-0.01	0.02	53.86	0.02

```r
# plot
p_binom2 = ggplot(raw_binom) +
  geom_boxplot(aes(x = Estimator,
                   y= Estimate),
               width = 0.1) +
  ggtitle("Binomial") +
  theme(axis.ticks.x = element_blank()) +
  facet_wrap(~ n)
```

- Normal

```r
set.seed(7272)

S1_norm <- list()
S2_norm <- list()

for(n in ss){
  S1 <- rep(NA, n.sims)
  S2 <- rep(NA, n.sims)
  for(i in 1:n.sims){
    curr.sample <- rnorm(n = n, mean = 0, sd = sqrt(2))
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_norm[[as.character(n)]] <- S1
  S2_norm[[as.character(n)]] <- S2
}
```

```r

# raw data
raw_norm <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_norm[["20"]], S2_norm[["20"]],
               S1_norm[["50"]], S2_norm[["50"]],
               S1_norm[["100"]], S2_norm[["100"]],
               S1_norm[["500"]], S2_norm[["500"]])
)
# summary
results_norm <- tibble(
  n = rep(c(20, 50, 100, 500), each=2),
  Estimator = rep(c("S1", "S2"), times = 4),
  Bias = c(
    mean(S1_norm[["20"]]) - 2, mean(S2_norm[["20"]]) - 2,
    mean(S1_norm[["50"]]) - 2, mean(S2_norm[["50"]]) - 2,
    mean(S1_norm[["100"]]) - 2, mean(S2_norm[["100"]]) - 2,
    mean(S1_norm[["500"]]) - 2, mean(S2_norm[["500"]]) - 2
  ),
  Variance = c(
    var(S1_norm[["20"]]),  var(S2_norm[["20"]]),
    var(S1_norm[["50"]]),  var(S2_norm[["50"]]),
    var(S1_norm[["100"]]), var(S2_norm[["100"]]),
    var(S1_norm[["500"]]), var(S2_norm[["500"]])
  ),
  Precision = 1 / Variance,
  MSE = Bias^2 + Variance)

# table
results_norm |> knitr::kable(digits = 2)
```

n	Estimator	Bias	Variance	Precision	MSE
20	S1	-0.01	0.41	2.41	0.41
20	S2	-0.11	0.37	2.67	0.39
50	S1	0.01	0.18	5.61	0.18
50	S2	-0.03	0.17	5.85	0.17
100	S1	0.00	0.09	11.36	0.09
100	S2	-0.02	0.09	11.59	0.09
500	S1	0.00	0.02	63.82	0.02
500	S2	0.00	0.02	64.07	0.02

```r
# plot
p_norm2 = ggplot(raw_norm) +
  geom_boxplot(aes(x = Estimator,
                   y= Estimate),
               width = 0.1) +
  ggtitle("Normal") +
  theme(axis.ticks.x = element_blank()) +
  facet_wrap(~ n)
```

- Exponential

```r
set.seed(7272)

```

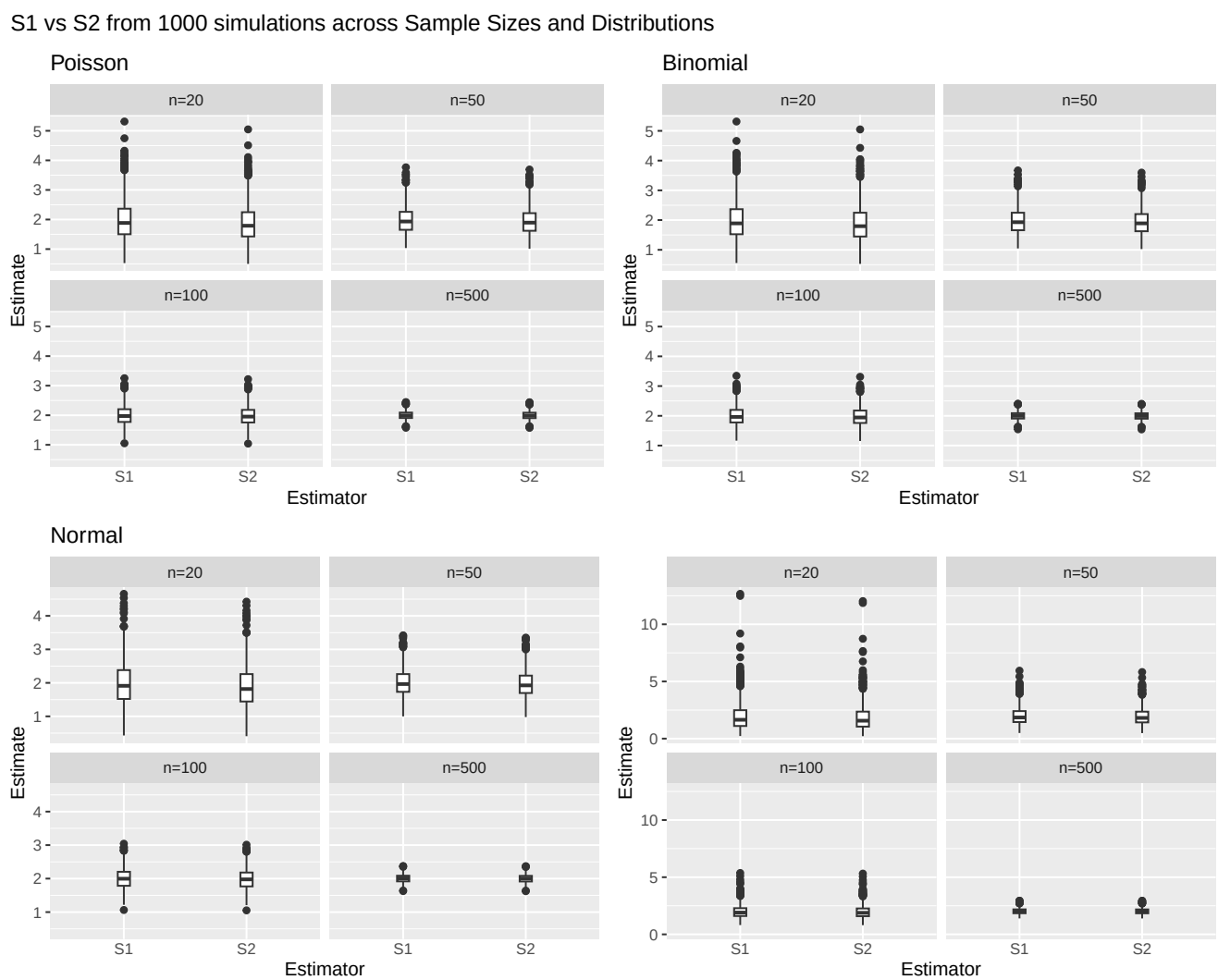
```r
S1_exp <- list()
S2_exp <- list()

for(n in ss){
  S1 <- rep(NA, n.sims)
  S2 <- rep(NA, n.sims)
  for(i in 1:n.sims){
    curr.sample <- rexp(n = n, rate = 1/sqrt(2))
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_exp[[as.character(n)]] <- S1
  S2_exp[[as.character(n)]] <- S2
}

# raw data
raw_exp <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_exp[["20"]], S2_exp[["20"]],
               S1_exp[["50"]], S2_exp[["50"]],
               S1_exp[["100"]], S2_exp[["100"]],
               S1_exp[["500"]], S2_exp[["500"]])
)
# summary
results_exp <- tibble(
  n = rep(c(20, 50, 100, 500), each=2),
  Estimator = rep(c("S1", "S2"), times = 4),
  Bias = c(
    mean(S1_exp[["20"]]) - 2, mean(S2_exp[["20"]]) - 2,
    mean(S1_exp[["50"]]) - 2, mean(S2_exp[["50"]]) - 2,
    mean(S1_exp[["100"]]) - 2, mean(S2_exp[["100"]]) - 2,
    mean(S1_exp[["500"]]) - 2, mean(S2_exp[["500"]]) - 2
  ),
  Variance = c(
    var(S1_exp[["20"]]),  var(S2_exp[["20"]]),
    var(S1_exp[["50"]]),  var(S2_exp[["50"]]),
    var(S1_exp[["100"]]), var(S2_exp[["100"]]),
    var(S1_exp[["500"]]), var(S2_exp[["500"]])
  ),
  Precision = 1 / Variance,
  MSE = Bias^2 + Variance)

# table
results_exp |> knitr::kable(digits = 2)
```

n	Estimator	Bias	Variance	Precision	MSE
20	S1	-0.03	1.64	0.61	1.64
20	S2	-0.12	1.48	0.68	1.49
50	S1	0.00	0.63	1.59	0.63
50	S2	-0.04	0.60	1.66	0.60
100	S1	-0.01	0.34	2.92	0.34
100	S2	-0.03	0.34	2.98	0.34
500	S1	0.02	0.06	15.93	0.06

n	Estimator	Bias	Variance	Precision	MSE
500	S2	0.02	0.06	16.00	0.06

```r
# plot
p_exp2 = ggplot(raw_exp) +
  geom_boxplot(aes(x = Estimator,
                   y= Estimate),
               width = 0.1) +
  ggtitle("") +
  theme(axis.ticks.x = element_blank()) +
  facet_wrap(~ n)
```

```r
(p_pois2 + p_binom2) / (p_norm2 + p_exp2) +
  plot_annotation(title = "S1 vs S2 from 1000 simulations across Sample Sizes and Distributions") +
  plot_layout(guides = "collect")
```



2.3 Simulation

You have written code to draw a random sample of various sizes from each distribution, and to calculate and store the calculated estimates. Rework your code to produce a density plot of the s^2_{n-1} and s^2_n. Do you have a guess about their distribution? Ensure to set your randomization seed.

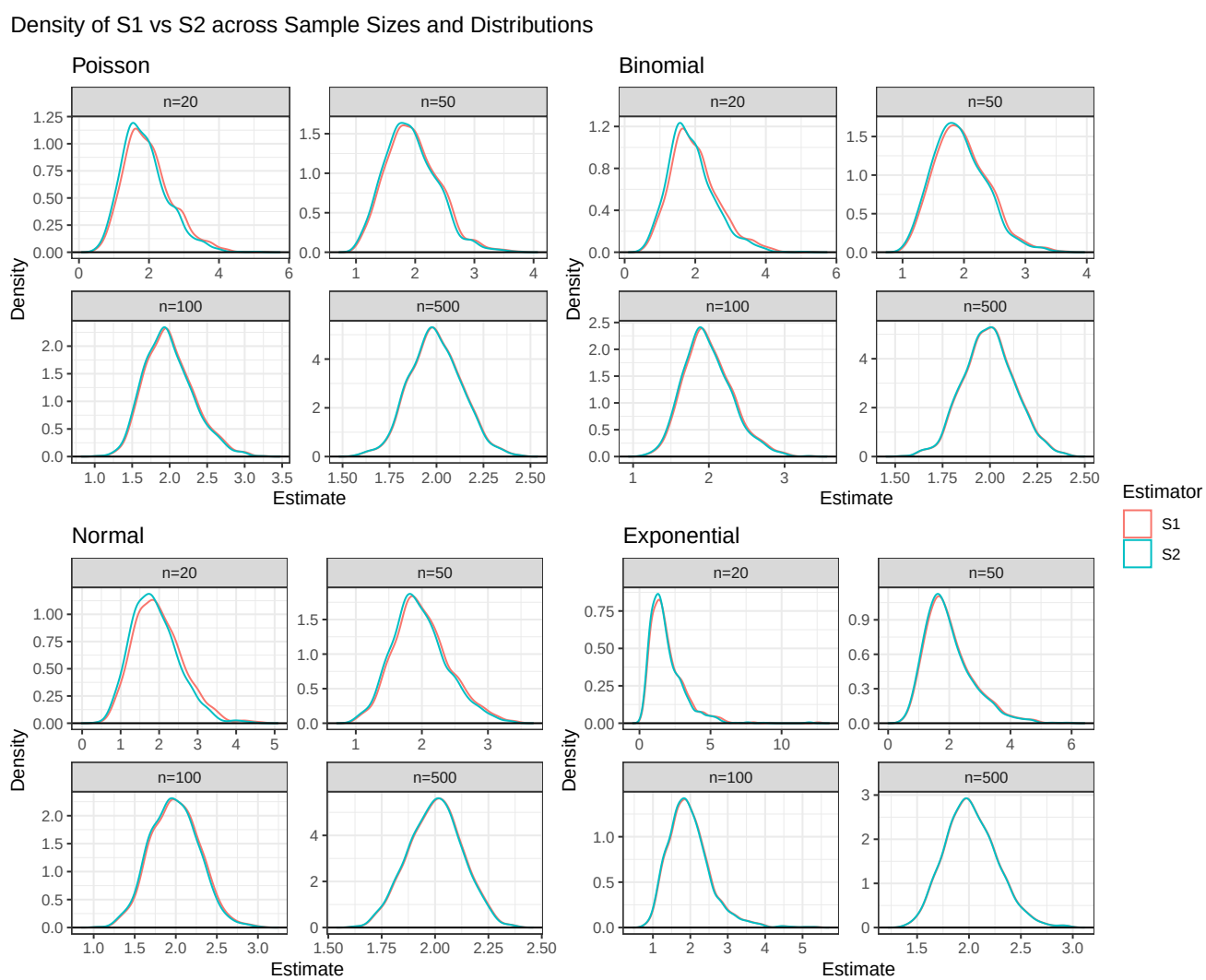
Solution

Based on the boxplots, their distributions become less right-skewed and have lower variance as sample size increases. When n=500, the distributions of S1 and S2 are approximately Gaussian.

```r
# n20.s1 <- tibble(
#   Poisson = raw_pois[raw_pois[raw_pois$n == "n=20" & raw_pois$Estimator == "S1"]],
#   Binomial = raw_binom[raw_binom[raw_binom$n == "n=20" & raw_binom$Estimator == "S1"]],
# )

set.seed(7272)
library(ggridges)

# raw_all <- bind_rows(
#   raw_pois |> mutate(Distribution = "Poisson"),
#   raw_binom |> mutate(Distribution = "Binomial"),
#   raw_norm |> mutate(Distribution = "Normal"),
#   raw_exp |> mutate(Distribution = "Exponential")
# )

# # S1
# raw_all |>
#   filter(Estimator == "S1") |>
#   ggplot(aes(x=Estimate, color = Distribution)) +
#   geom_density() +
#   facet_wrap(~ n) +
#   xlab("Estimate") +
#   ylab("Density") +
#   ggtitle("Sampling Distribution of S1 across sample sizes and distributions")

# poisson
# raw_pois |>
#       filter(Estimator == "S1") |>
#       ggplot(aes(x=Estimate, color = n)) +
#       geom_density() +
#       xlab("S1") +
#       ylab("Density") +
#       ggtitle("Poisson")

# poisson

p_pois1 = ggplot(raw_pois) +
  geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Poisson") +
  # lets each facet panel have its own x and y axes instead of forcing them to share the same range
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()

# binomial
p_binom1 = ggplot(raw_binom) +
  geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Binomial") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

```r

# normal
p_norm1 = ggplot(raw_norm) +
  geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Normal") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()

# Exponential
p_exp1 = ggplot(raw_exp) +
  geom_density_ridges(aes(x = Estimate, y=0, color=`Estimator`), fill=NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Exponential") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

```r
(p_pois1 + p_binom1) / (p_norm1 + p_exp1) +
  plot_annotation(title = "Density of S1 vs S2 across Sample Sizes and Distributions") +
  plot_layout(guides = "collect")
```



2.4 Bootstrap

Now, take *one* sample from each distribution. Instead of drawing a random sample at each iteration, sample *with replacement* from the one sample. Produce a density plot of the s^2_{n-1} and s^2_n. Do we get roughly the same information as we do in the full simulation? Try this over several seeds – what do you notice? Ensure to set your randomization seed.

Solution

The distributions of S1 and S2 across distributions resemble a Gaussian distribution, even when the sample size is small. They do become more

- Poisson

```r
set.seed(7272)

# poisson
one_pois <- list()
S1_pois_b <- list()
S2_pois_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  # take 1 random sample of different sizes
  one_pois[[as.character(n)]] <- rpois(n = n, lambda = 2)
  for(i in 1:n.sims){
    curr.sample <- sample(one_pois[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_pois_b[[as.character(n)]] <- S1
  S2_pois_b[[as.character(n)]] <- S2
}

raw_b_pois <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_pois_b[["20"]], S2_pois_b[["20"]],
               S1_pois_b[["50"]], S2_pois_b[["50"]],
               S1_pois_b[["100"]], S2_pois_b[["100"]],
               S1_pois_b[["500"]], S2_pois_b[["500"]])
)

p_pois = ggplot(raw_b_pois) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Poisson") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

- Binomial

```r
set.seed(7272)

one_binom <- list()
S1_binom_b <- list()
```

```r
S2_binom_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_binom[[as.character(n)]] <- rbinom(n = n, size = 50, prob = p)
  for(i in 1:n.sims){
    curr.sample <- sample(one_binom[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_binom_b[[as.character(n)]] <- S1
  S2_binom_b[[as.character(n)]] <- S2
}

raw_b_binom <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_binom_b[["20"]], S2_binom_b[["20"]],
               S1_binom_b[["50"]], S2_binom_b[["50"]],
               S1_binom_b[["100"]], S2_binom_b[["100"]],
               S1_binom_b[["500"]], S2_binom_b[["500"]])
)

p_binom = ggplot(raw_b_binom) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Binomial") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

- Normal

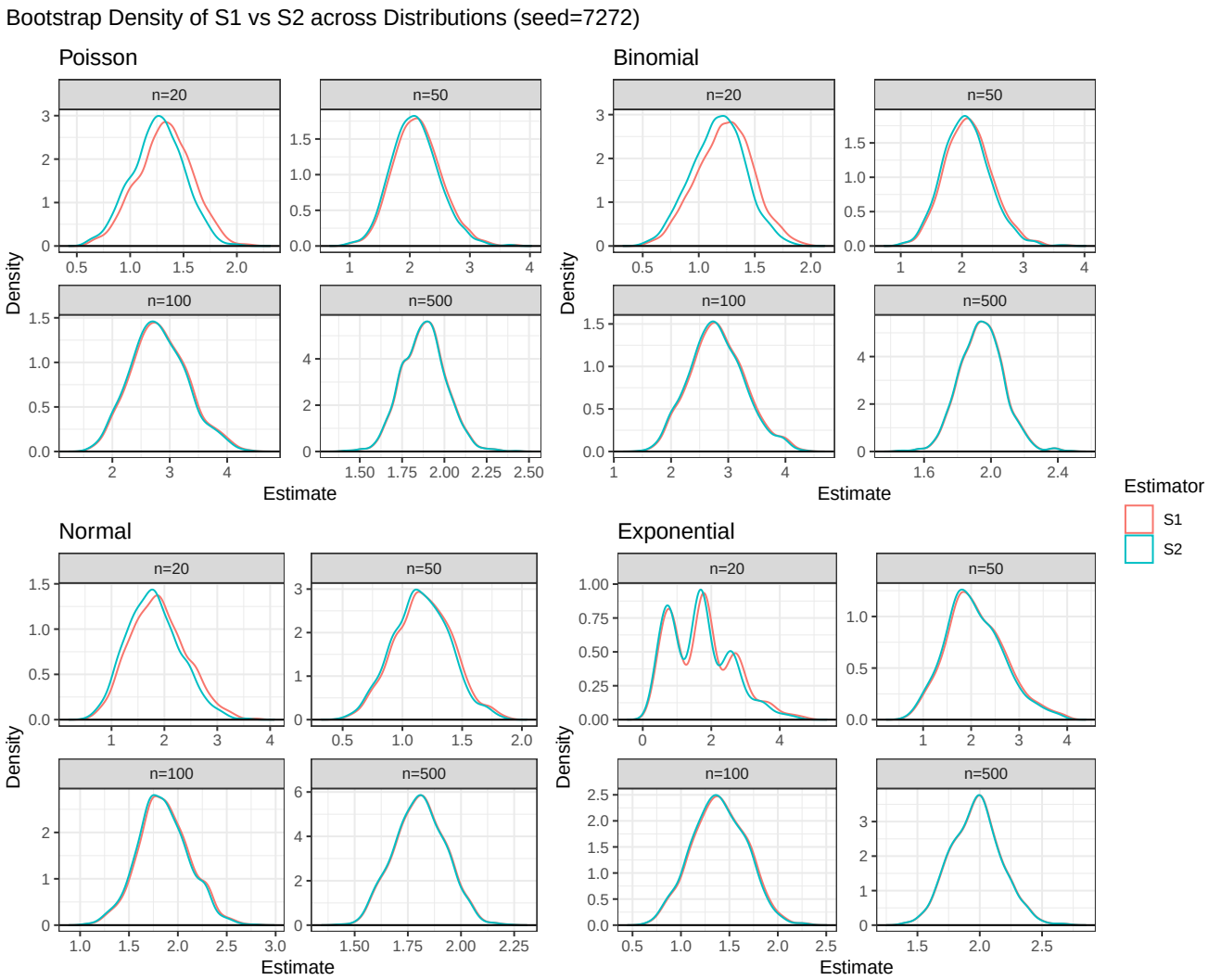
```r
set.seed(7272)

one_norm <- list()
S1_norm_b <- list()
S2_norm_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_norm[[as.character(n)]] <- rnorm(n = n, mean = 0, sd = sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_norm[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_norm_b[[as.character(n)]] <- S1
  S2_norm_b[[as.character(n)]] <- S2
}

raw_b_norm <- tibble(
```

```r
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_norm_b[["20"]], S2_norm_b[["20"]],
               S1_norm_b[["50"]], S2_norm_b[["50"]],
               S1_norm_b[["100"]], S2_norm_b[["100"]],
               S1_norm_b[["500"]], S2_norm_b[["500"]])
)

p_norm = ggplot(raw_b_norm) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Normal") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

- Exponential

```r
set.seed(7272)

one_exp <- list()
S1_exp_b <- list()
S2_exp_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_exp[[as.character(n)]] <- rexp(n = n, rate = 1/sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_exp[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_exp_b[[as.character(n)]] <- S1
  S2_exp_b[[as.character(n)]] <- S2
}

raw_b_exp <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_exp_b[["20"]], S2_exp_b[["20"]],
               S1_exp_b[["50"]], S2_exp_b[["50"]],
               S1_exp_b[["100"]], S2_exp_b[["100"]],
               S1_exp_b[["500"]], S2_exp_b[["500"]])
)

p_exp = ggplot(raw_b_exp) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Exponential") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```

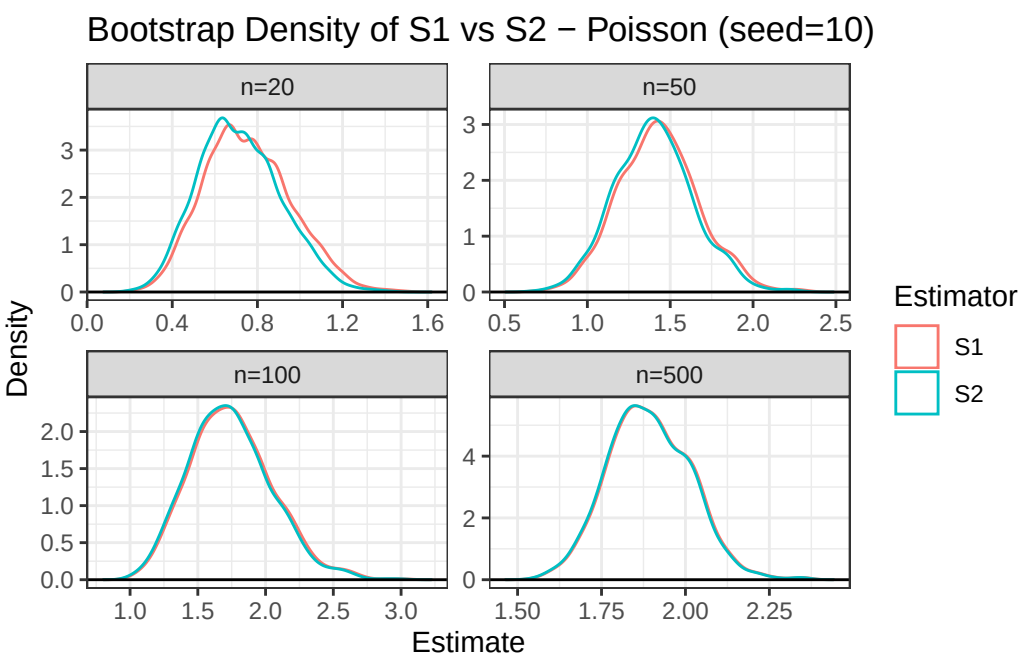
```r
(p_pois + p_binom) / (p_norm + p_exp) +
  plot_annotation(title = "Bootstrap Density of S1 vs S2 across Distributions (seed=7272)") +
  plot_layout(guides = "collect")
```



- Trying this with seed = 10

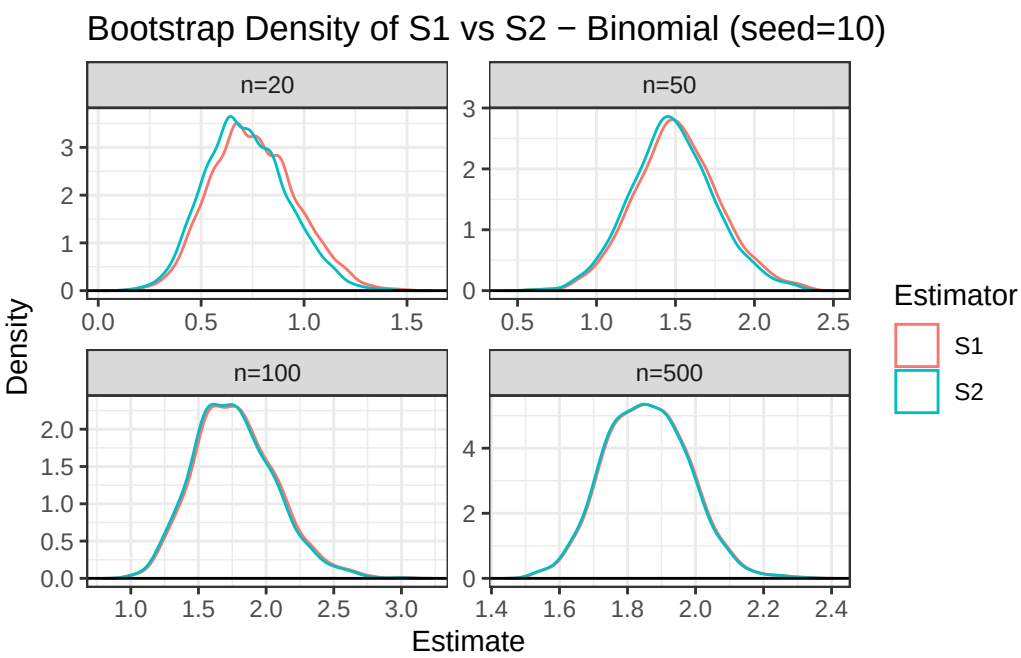
```r
set.seed(10)

# poisson
one_pois <- list()
S1_pois_b <- list()
S2_pois_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  # take 1 random sample of different sizes
  one_pois[[as.character(n)]] <- rpois(n = n, lambda = 2)
  for(i in 1:n.sims){
    curr.sample <- sample(one_pois[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
```

```r
  }
  S1_pois_b[[as.character(n)]] <- S1
  S2_pois_b[[as.character(n)]] <- S2
}

raw_b_pois <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_pois_b[["20"]], S2_pois_b[["20"]],
               S1_pois_b[["50"]], S2_pois_b[["50"]],
               S1_pois_b[["100"]], S2_pois_b[["100"]],
               S1_pois_b[["500"]], S2_pois_b[["500"]])
)

ggplot(raw_b_pois) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Poisson (seed=10)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



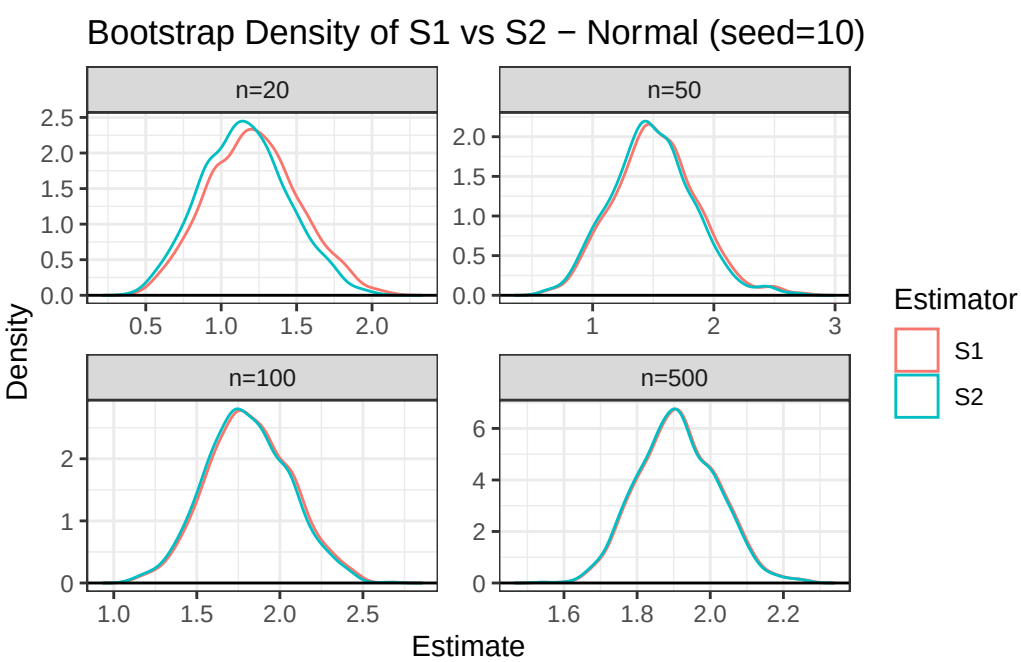

```r
set.seed(10)

one_binom <- list()
S1_binom_b <- list()
S2_binom_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_binom[[as.character(n)]] <- rbinom(n = n, size = 50, prob = p)
  for(i in 1:n.sims){
    curr.sample <- sample(one_binom[[as.character(n)]], size = n, replace = TRUE)
```

```r
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_binom_b[[as.character(n)]] <- S1
  S2_binom_b[[as.character(n)]] <- S2
}

raw_b_binom <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_binom_b[["20"]], S2_binom_b[["20"]],
               S1_binom_b[["50"]], S2_binom_b[["50"]],
               S1_binom_b[["100"]], S2_binom_b[["100"]],
               S1_binom_b[["500"]], S2_binom_b[["500"]])
)

ggplot(raw_b_binom) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Binomial (seed=10)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



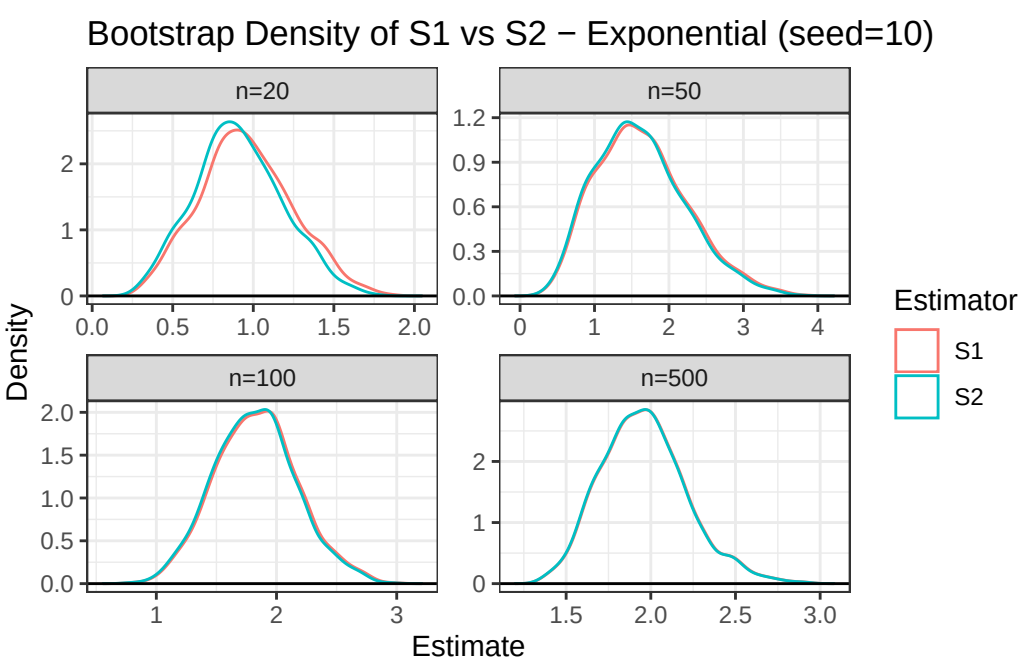

```r
set.seed(10)

one_norm <- list()
S1_norm_b <- list()
S2_norm_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
```

```r
  one_norm[[as.character(n)]] <- rnorm(n = n, mean = 0, sd = sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_norm[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_norm_b[[as.character(n)]] <- S1
  S2_norm_b[[as.character(n)]] <- S2
}

raw_b_norm <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_norm_b[["20"]], S2_norm_b[["20"]],
               S1_norm_b[["50"]], S2_norm_b[["50"]],
               S1_norm_b[["100"]], S2_norm_b[["100"]],
               S1_norm_b[["500"]], S2_norm_b[["500"]])
)

ggplot(raw_b_norm) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Normal (seed=10)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



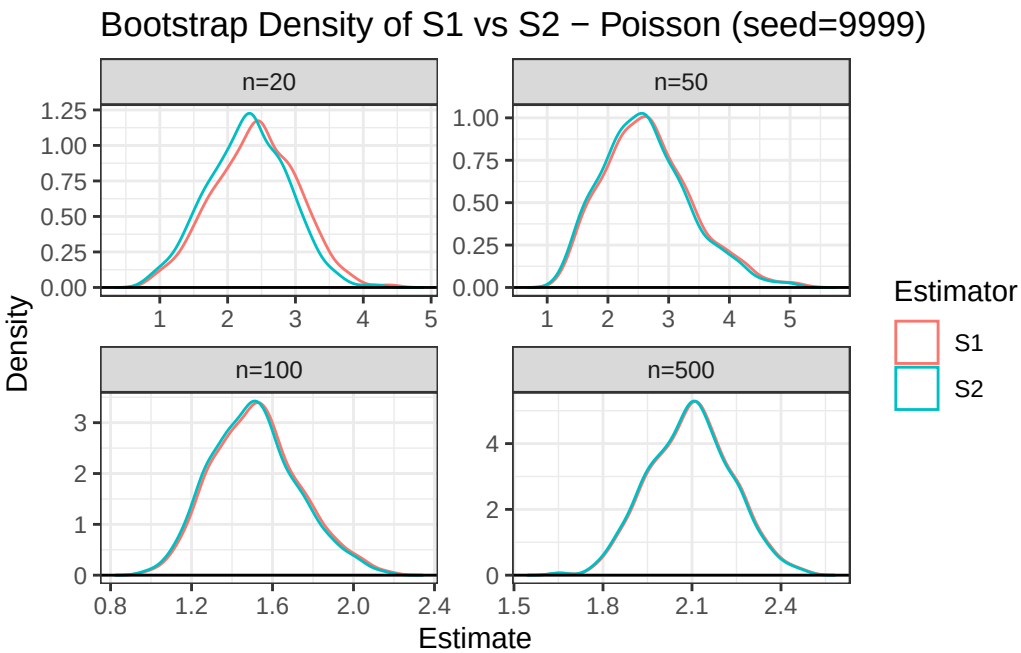

```r
set.seed(10)

one_exp <- list()
S1_exp_b <- list()
S2_exp_b <- list()
S1 <- rep(NA, n.sims)
```

```r
S2 <- rep(NA, n.sims)

for(n in ss){
  one_exp[[as.character(n)]] <- rexp(n = n, rate = 1/sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_exp[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_exp_b[[as.character(n)]] <- S1
  S2_exp_b[[as.character(n)]] <- S2
}

raw_b_exp <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_exp_b[["20"]], S2_exp_b[["20"]],
               S1_exp_b[["50"]], S2_exp_b[["50"]],
               S1_exp_b[["100"]], S2_exp_b[["100"]],
               S1_exp_b[["500"]], S2_exp_b[["500"]])
)

ggplot(raw_b_exp) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Exponential (seed=10)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



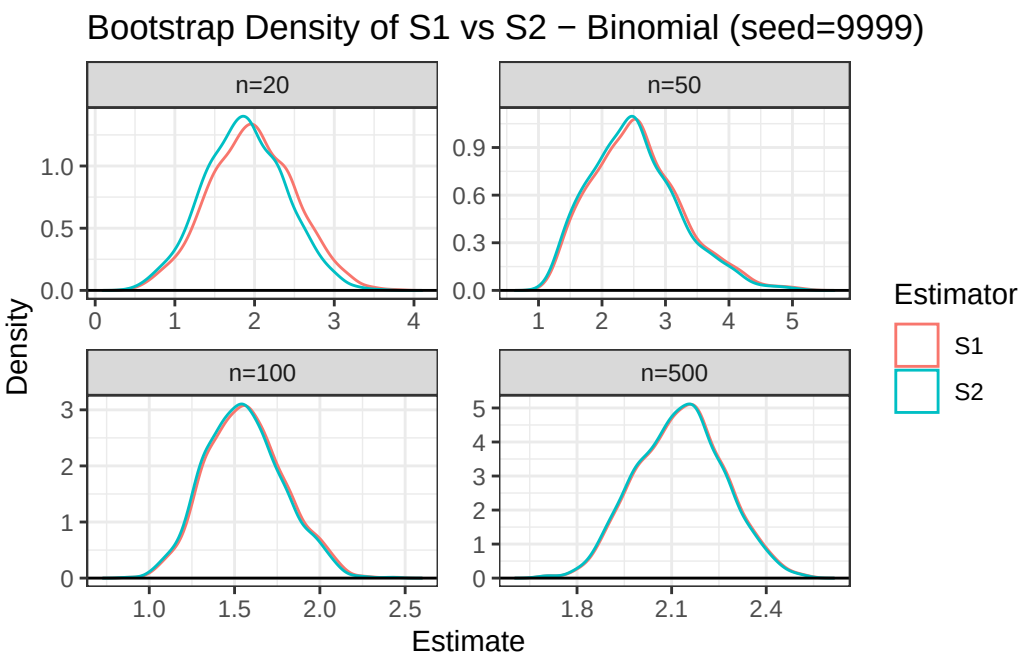
- Trying this with seed = 99999

```r
set.seed(99999)

# poisson
one_pois <- list()
S1_pois_b <- list()
S2_pois_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  # take 1 random sample of different sizes
  one_pois[[as.character(n)]] <- rpois(n = n, lambda = 2)
  for(i in 1:n.sims){
    curr.sample <- sample(one_pois[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_pois_b[[as.character(n)]] <- S1
  S2_pois_b[[as.character(n)]] <- S2
}

raw_b_pois <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_pois_b[["20"]], S2_pois_b[["20"]],
               S1_pois_b[["50"]], S2_pois_b[["50"]],
               S1_pois_b[["100"]], S2_pois_b[["100"]],
               S1_pois_b[["500"]], S2_pois_b[["500"]])
)

ggplot(raw_b_pois) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Poisson (seed=9999)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



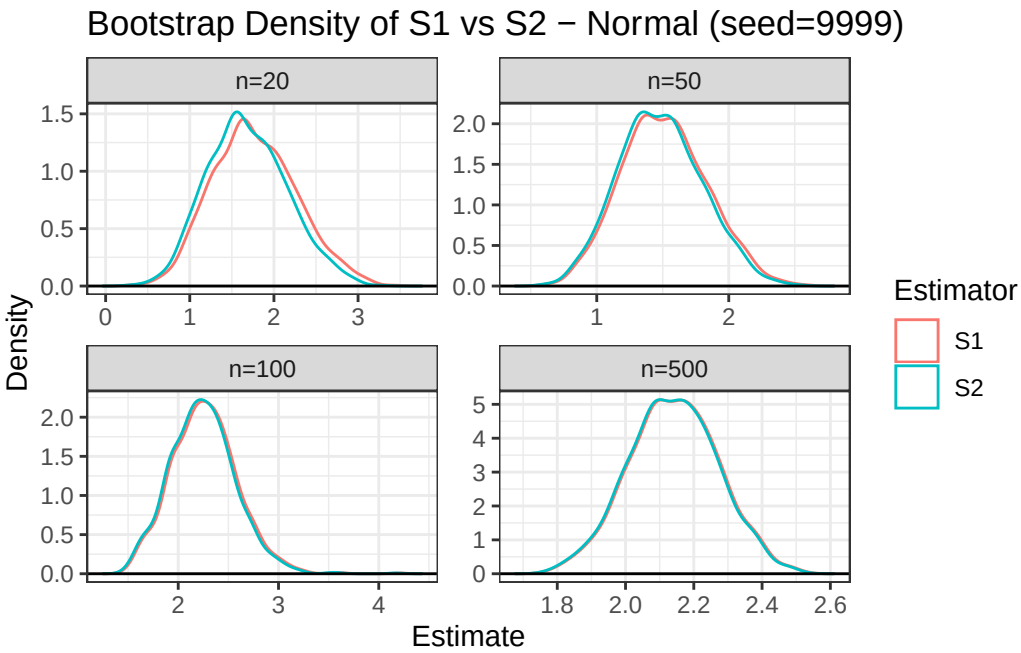

```r
set.seed(99999)

one_binom <- list()
S1_binom_b <- list()
S2_binom_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_binom[[as.character(n)]] <- rbinom(n = n, size = 50, prob = p)
  for(i in 1:n.sims){
    curr.sample <- sample(one_binom[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_binom_b[[as.character(n)]] <- S1
  S2_binom_b[[as.character(n)]] <- S2
}

raw_b_binom <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_binom_b[["20"]], S2_binom_b[["20"]],
               S1_binom_b[["50"]], S2_binom_b[["50"]],
               S1_binom_b[["100"]], S2_binom_b[["100"]],
               S1_binom_b[["500"]], S2_binom_b[["500"]])
)

ggplot(raw_b_binom) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Binomial (seed=9999)") +
  facet_wrap(~ n, scales = "free") +
```

```r
  ylab("Density") +
  theme_bw()
```



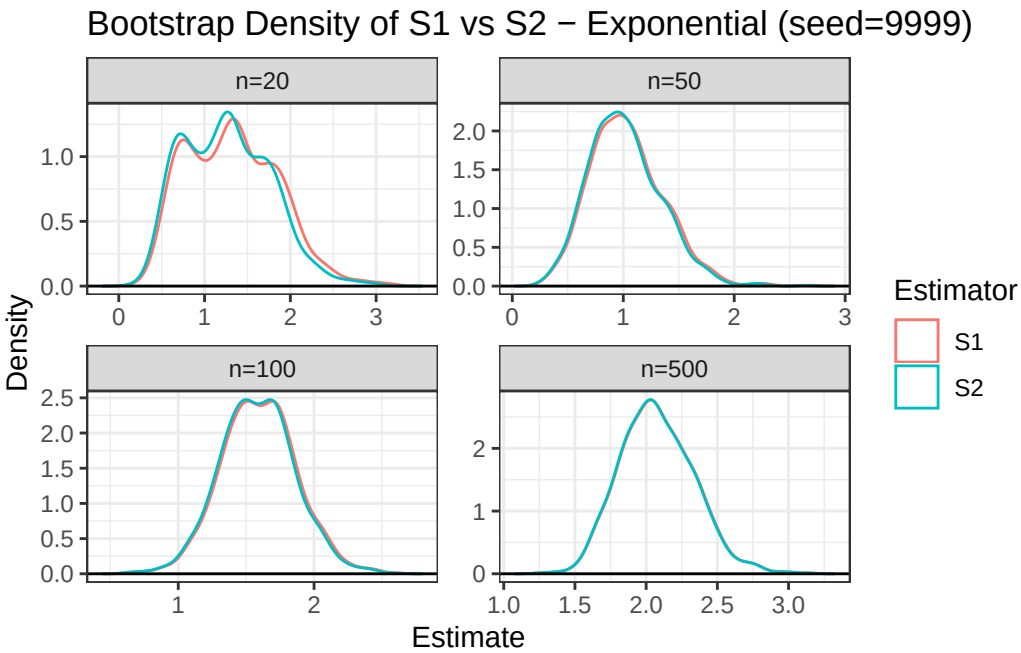

```r
set.seed(99999)

one_norm <- list()
S1_norm_b <- list()
S2_norm_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_norm[[as.character(n)]] <- rnorm(n = n, mean = 0, sd = sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_norm[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_norm_b[[as.character(n)]] <- S1
  S2_norm_b[[as.character(n)]] <- S2
}

raw_b_norm <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_norm_b[["20"]], S2_norm_b[["20"]],
               S1_norm_b[["50"]], S2_norm_b[["50"]],
               S1_norm_b[["100"]], S2_norm_b[["100"]],
               S1_norm_b[["500"]], S2_norm_b[["500"]])
)

ggplot(raw_b_norm) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
```

```r
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Normal (seed=9999)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```




```r
set.seed(99999)

one_exp <- list()
S1_exp_b <- list()
S2_exp_b <- list()
S1 <- rep(NA, n.sims)
S2 <- rep(NA, n.sims)

for(n in ss){
  one_exp[[as.character(n)]] <- rexp(n = n, rate = 1/sqrt(2))
  for(i in 1:n.sims){
    curr.sample <- sample(one_exp[[as.character(n)]], size = n, replace = TRUE)
    x_bar <- mean(curr.sample)
    S1[i] <- sum((curr.sample - x_bar)^2) / (n - 1)
    S2[i] <- sum((curr.sample - x_bar)^2) / n
  }
  S1_exp_b[[as.character(n)]] <- S1
  S2_exp_b[[as.character(n)]] <- S2
}

raw_b_exp <- tibble(
  Estimator = rep(c("S1", "S2"), each = 1000, times = 4),
  n = factor(rep(c("n=20", "n=50", "n=100", "n=500"), each = 2000),
             levels = c("n=20", "n=50", "n=100", "n=500")),
  Estimate = c(S1_exp_b[["20"]], S2_exp_b[["20"]],
               S1_exp_b[["50"]], S2_exp_b[["50"]],
               S1_exp_b[["100"]], S2_exp_b[["100"]],
               S1_exp_b[["500"]], S2_exp_b[["500"]])
)
```

```r

ggplot(raw_b_exp) +
  geom_density_ridges(aes(x = Estimate, y = 0, color = Estimator), fill = NA) +
  geom_hline(yintercept = 0) +
  ggtitle("Bootstrap Density of S1 vs S2 - Exponential (seed=9999)") +
  facet_wrap(~ n, scales = "free") +
  ylab("Density") +
  theme_bw()
```



3 Executive Summary

Summarize the work you've done here to provide a brief (200-300 word) summary of the simulation and the results using your work in Lab 11. Your summary should be professional, clear, and all claims should be corroborated with statistical support.